

Large-scale photonic computing with nonlinear disordered media

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Neural networks find widespread use in scientific and technological applications, yet their implementations in conventional computers have encountered bottlenecks due to ever-expanding computational needs. Photonic computing is a promising neuromorphic platform with potential advantages of massive parallelism, ultralow latency and reduced energy consumption but mostly for computing linear operations. Here we demonstrate a large-scale, high-performance nonlinear photonic neural system based on a disordered polycrystalline slab composed of lithium niobate nanocrystals. Mediated by random quasi-phase-matching and multiple scattering, linear and nonlinear optical speckle features are generated as the interplay between the simultaneous linear random scattering and the second-harmonic generation, defining a complex neural network in which the second-order nonlinearity acts as internal nonlinear activation functions. Benchmarked against linear random projection, such nonlinear mapping embedded with rich physical computational operations shows improved performance across a large collection of machine learning tasks in image classification, regression and graph classification. Demonstrating up to 27,648 input and 3,500 nonlinear output nodes, the combination of optical nonlinearity and random scattering serves as a scalable computing engine for diverse applications.

From combinatorial optimization algorithms that exhaustively search solutions of real-world problems to artificial neural networks (NNs) that support substantial technology revolutions, modern digital computers have been excellent hardware and major workhorse for decades. However, they recently appear struggling to keep pace with the booming trend of operation-dense applications by consuming far too much energy, memory and time for data management, storage, training and inference. The main bottleneck of contemporary digital computers can be attributed to the architecture they are built upon, the von Neumann

architecture, where the memory and the central processing unit are physically separated and information is processed sequentially. A fundamentally different path in computation is to bring the memory or signal to the processing units as close as possible, thereby achieving speed and efficiency advantages. There has been a continuous drive to develop NN accelerators that employ various technologies, from electronics¹, mechanical² and photonic neuromorphic hardware^{3,4} to memristor⁵ and DNA computing devices⁶. All of these endeavors are in a shared wish to extend brain-inspired information processing to

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physical systems, for more scalable, environment-friendly and even capacity-enhanced machine learning (ML)⁷.

Photonic neural systems are one of the promising platforms for analog computing that show prospects for massive parallelism, ultralow latency and reduced power consumption^{3,4}, with a rich history of exploration dating back to the Hopfield network⁸ and optical correlators⁹ in the past century. Recently, optical computing has been revived following the rapid advancements of deep NNs. Various distinct matrix–vector multipliers have been implemented in photonic chips for inference^{10–14}. Free-space light propagation through linear diffractive¹⁵ or convolutional layers^{16,17} has also been leveraged to execute large-scale multiplications. Apart from photonic systems that are isomorphism of digital NNs, computation is also naturally performed when light scatters through disordered optical media¹⁸. Feeding the system with, for example, spatially modulated light, multiple scattering computes random projection (RP)—a ubiquitous operation that is broadly used in mathematics and statistics. The optical RP can be of an extreme large scale, and its random matrix is physically stored in the medium itself instead of auxiliary memory. Kernel ridge regression can leverage such large-scale computations for ML applications¹⁹. Similar to reservoir computing and its photonic implementations^{20–26}, they are easy to train with only a readout layer at the output, while leaving the optical part, where most of the computations are performed, untrained. However, multiple light scattering is mostly linear in nature even under high-power or pulsed illumination, thus making the process equivalent to a single-layer NN. Similar to digital NNs, nonlinear activation functions are indispensable to obtain a deep photonic neural system with appreciable expressivity. Several methods to realize nonlinear functions have been developed, from square-law detector nonlinearity^{15,25,27}, atom-induced nonlinearity^{28,29} and phase-change nonlinearity³⁰ to electrooptic, optoelectronic and optoelectrooptic nonlinearities^{20,31–34}. Yet, the vast majority of these nonlinearities are either incoherent or convert the optical waves to other formats, thereby posing challenges for subsequent photonic processing. Noticeably, the utilization of coherent optical nonlinearity was envisioned three decades ago^{35,36} and has recently been revived^{37–39} and demonstrated in nonlinear spatiotemporal evolution of optical pulses in multimode fibers⁴⁰. Leveraging the third-order ($\chi^{(3)}$) nonlinearity of a silica fiber, the nonlinear photonic neural system is demonstrated up to a few hundreds of nodes, limited by the number of modes supported by the waveguide.

Unlike the $\chi^{(3)}$ nonlinearity that is ubiquitous across all material platforms, the second-order ($\chi^{(2)}$) nonlinearity is only present in non-centrosymmetric media. As a lower-order nonlinearity, $\chi^{(2)}$ nonlinear effects are more efficient provided with proper phase matching, hence it is also interesting for nonlinear optical computation. In this Article, we demonstrate a large-scale photonic NN that combines linear scattering and $\chi^{(2)}$ optical nonlinearity for a wide range of ML applications. The core processing unit consists in a disordered polycrystalline lithium niobate (LN) slab assembled from nanocrystals⁴¹, which not only is multiple scattering but also generates second-harmonic (SH) light assisted by random quasi-phase-matching⁴². The linear scattering of both fundamental-harmonic (FH) and SH waves contributes to fixed random weights, while the optical nonlinearity contributes to activation functions, defining a complex photonic NN isomorphic to a massive third-order scattering tensor⁴³. We encode the input information to the optical wavefront of the FH and record the speckle intensity of the SH at the output of the LN slab. As such, nonlinear features are extracted from the SH speckle and are then fed into a simple linear regression layer to solve ML tasks. Meanwhile, the optical system also naturally computes the linear speckle at the FH as the experimental baseline⁴⁹. Compared with linear speckle features together with the simulated RP results, we experimentally verify that the nonlinear features exhibit improved representational capabilities though showing different gains across 14 diverse datasets of varying complexity in image classification, univariate and multivariate regression and graph classification. These results

unequivocally identify the role of optical nonlinearity in large-scale complex photonic NNs, with the number of input and nonlinear output nodes up to 27,648 and 3,500, respectively. The nonlinear photonic neural system demonstrated here distinguishes itself from previous photonic reservoir computing schemes^{21–26}, owing to its inherent optical nonlinearity and richer computations.

Results

Figure 1a shows the schematic setup of our photonic neural system, comprising data encoding, optical feature extraction, intensity detection and digital readout. First, we encode the input data to the spatial phase profile of the FH pulse at a central wavelength of 800 nm (see Methods for experimental setup). Noticeably, phase encoding using a spatial light modulator (SLM) inherently introduces element-wise nonlinearity⁴⁴, which transforms the input data x to the input field $E_{\text{in}} = \exp(i\pi x)$. Then, the modulated FH wave illuminates the disordered LN slab to generate optical features (see Methods for sample information), thereby projecting input data into a high-dimensional latent space. The feature extraction includes both linear and nonlinear parts: in addition to the linear scattering of the FH wave, the SH at 400 nm is generated mediated by random quasi-phase-matching and scattered in the meantime⁴¹. The intensity of both the linear and nonlinear speckle fields are captured by their respective cameras, forming linear and nonlinear speckle features. For either FH or SH feature extraction, a computationally-efficient digital linear regression layer is trained to perform various ML tasks (see Methods for data processing). Note that we employ respective digital linear layers with identical dimensions to ensure a fair comparison between the two feature extraction processes. In this work, we designate the linear and nonlinear features based on their wavelengths, as our primary focus is on the role of optical nonlinearity instead of phase encoding and intensity detection nonlinearity, which are identical in both cases. The overall workflow is summarized in Supplementary Fig. 1.

To explain the linear and nonlinear feature computations in detail, we illustrate in Fig. 1b,c their physical and approximated mathematical forward models (Supplementary Note 1). The FH speckle pertains to a classical multiply scattering process in optics, recently exploited to compute RP¹⁸. Although the ultrashort FH pulse used in our experiment corresponds to a few spectral correlation bandwidth of the medium⁴⁵ (Supplementary Fig. 2), the linear scattering process can be adequately approximated by a single transmission matrix $W_l \in \mathbb{C}^{N \times M}$ (ref. 46), connecting the input field $E_{\text{in}} \in \mathbb{C}^{M \times 1}$ and output field $E_{\text{out,FH}} \in \mathbb{C}^{N \times 1}$ by $E_{\text{out,FH}} = W_l E_{\text{in}}$, where M and N are the dimensions of input and output, respectively. Indeed, the computing results based on our linear speckle features obtained with pulsed excitation are very similar to RP achieved using continuous-wave (CW) light¹⁹.

The formation of the SH speckle involves more intricate physical processes. To conceptualize it, we imagine the LN slab is sliced into multiple thin units, with each unit comprising several LN nanocubes (Fig. 1c). At the input side of a specific unit, we have the FH and generated SH from the previous unit. These incoming FH and SH waves are weakly scattered inside the current unit, described by some fixed and linear connections. Meanwhile, a portion of the FH is nonlinearly transformed into the SH and add on top on the current SH wave. Consequently, the scattered SH fields formed at the camera plane represent the interference of the generated SH from all the units. Collectively, the physical interconnection between the input field $E_{\text{in}} \in \mathbb{C}^{M \times 1}$ and nonlinear output field $E_{\text{out,FH}} \in \mathbb{C}^{N \times 1}$ can be accurately modeled as $E_{\text{out,SH}} = W_n \text{vec}(E_{\text{in}} \otimes E_{\text{in}})$, where $\otimes$ and 'vec()' denote the outer product operation and the vectorization of a matrix, respectively. $W_n \in \mathbb{C}^{N \times M^2}$ is a matrix reshaped from a third-order scattering tensor that fully characterizes the nonlinear scattering process as demonstrated recently⁴³. In comparison, duplicating the input by M copies in the FH case (to form a matrix of size $N \times M^2$) is not equivalent to the tensor computation in the SH case, simply due to its linear nature.

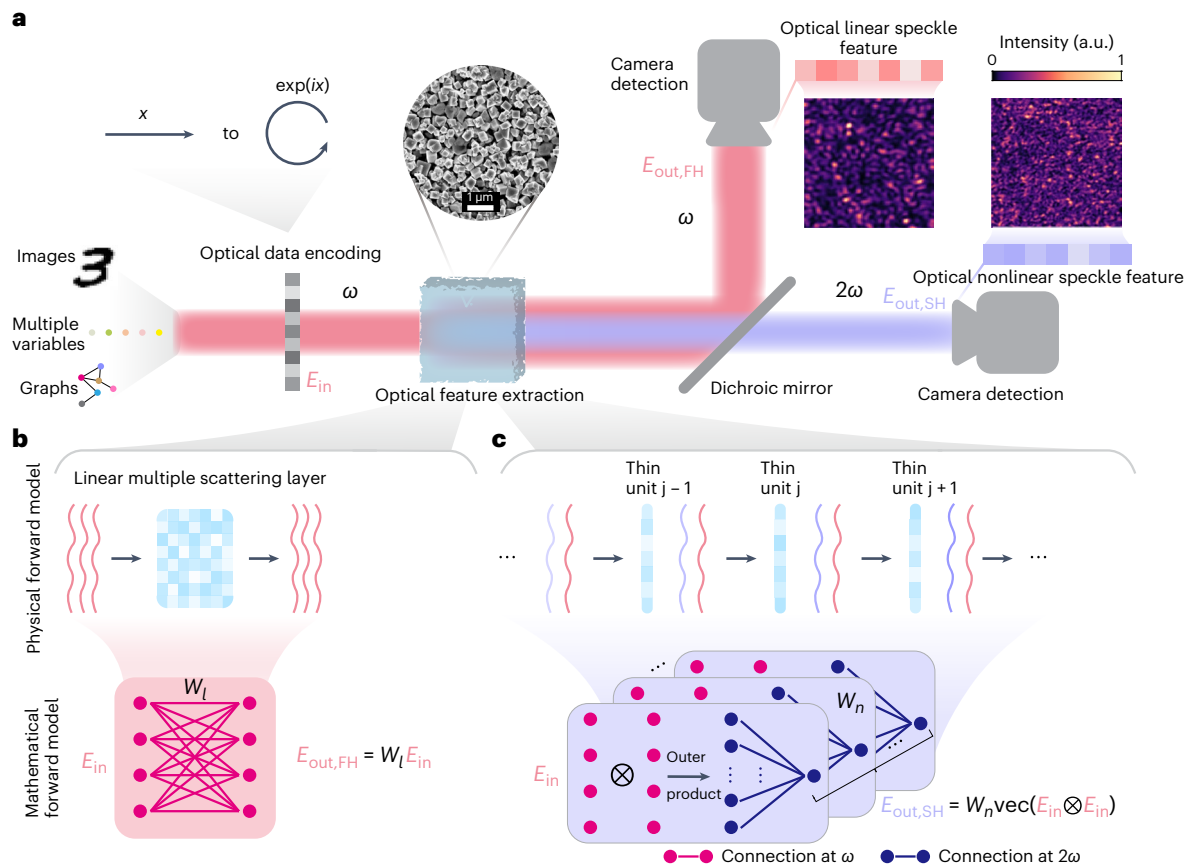


Fig. 1 | Optical feature extraction and forward models. a, The schematic setup for linear and nonlinear feature extraction. A pulsed laser at the FH illuminates a disordered scattering LN slab (a scanning electron microscope image is shown) and generates FH (ω , red) and SH (2ω , violet) speckle features, which are separated by a dichroic mirror and measured by cameras. The input data of various formats are encoded onto the spatial phase of light (denoted by x to $\exp(ix)$) and are then transformed to the intensity of the FH and SH speckles (their representative measured speckle patterns are shown, respectively). The speckle vectors are then fed into a linear digital readout layer to make task decisions. **b**, The physical and mathematical forward models of FH speckle features. The linear propagation in a complex medium can be mathematically

described as a complex-valued random matrix $W_l \in \mathbb{C}^{N \times M}$, linearly connecting the input field $E_{in} \in \mathbb{C}^{M \times 1}$ and the output field $E_{out,FH} \in \mathbb{C}^{N \times 1}$. **c**, The physical and mathematical forward models of SH speckle features. The SH speckle generation process can be dissected by segmenting the LN slab into thin units. In each unit, FH and SH waves experience weak scattering and a portion of the FH is nonlinearly transformed to SH. Overall, the process can be mathematically described by a more complex nonlinear mapping $E_{out,SH} = W_n \text{vec}(E_{in} \otimes E_{in})$, where $W_n \in \mathbb{C}^{N \times M^2}$ is a matrix reshaped from a third-order tensor that connects the input field $E_{in} \in \mathbb{C}^{M \times 1}$ and nonlinear output field $E_{out,SH} \in \mathbb{C}^{N \times 1}$. $\otimes$, outer product; $\text{vec}()$, matrix vectorization. Each output node of $E_{out,SH}$ (violet dot) is nonlinearly connected to input nodes (one block in W_n).

Image classification

To assess the added value of nonlinearity, we begin by applying our photonic neural system to image classification. Figure 2 showcases the efficacy of linear and nonlinear optical features for three image classification tasks with varying complexity. The first task is to classify 10-class sign language digits (SLD; Fig. 2a–d)⁴⁷. We encode the SLD (64×64 pixels) to the SLM, with each pixel from the image corresponding to a macropixel at SLM (see Methods and Supplementary Table 1 for data processing). Then, we record the speckle features with cameras and use them for digit classification. Figure 2b,c shows the confusion matrices obtained separately by 3,500 nonlinear and linear optical feature nodes, with test accuracies of 85.96% and 74.82% via respective linear digital fully connected layers of dimension $3,500 \times 10$. To validate whether the nonlinear advantage is concrete and universal, we search the optimal Tikhonov regularization parameters for both linear and nonlinear speckle features at different feature sizes and also simulate the linear RP as the benchmark shown in Fig. 2d. All of them are marked with one standard deviation (s.d.) from ten repeated trials for each case. The experimental linear features and simulated RP show very similar classification performance, while the experimental nonlinear features consistently outperform both of them. At various feature sizes, utilizing the nonlinear features leads to average accuracy

improvements ranging from 5.4% to 10.9% compared with using their linear counterparts.

Then, we proceed to the American Sign Language (ASL) alphabet dataset (Fig. 2e–h)⁴⁸, which consists of 27,455 training and 7,172 test images that represent 24 English letters (except ‘J’ and ‘Z’). Nonlinear speckles show as more expressive features with a test accuracy of 83.51%, compared with 78.53% obtained with linear speckles. An average accuracy improvement of 4.6% is obtained for feature sizes ranging from 500 to 3,500. Notably, such an accuracy gap may not be easily filled by increasing the dimension of linear features, as the linear features will eventually approach certain kernel limit with a performance upper bound⁴⁹. This also reflects the unique role of optical nonlinearity added to the RP. Additionally, we test our photonic neural system for commonly used Modified National Institute of Standards and Technology (MNIST)⁴⁹ and FashionMNIST⁵⁰ datasets (Supplementary Fig. 3). For these tasks, linear optical RP is already a quite effective approach to extract features, as evident by the classification accuracy compared with relevant works^{2,51,52}, but we still observe a performance improvement using nonlinear speckle features (Supplementary Table 2).

Moreover, we challenge our photonic neural system with more demanding tasks. Figure 2i–l demonstrates the experimental results for the Canadian Institute For Advanced Research (CIFAR-10)

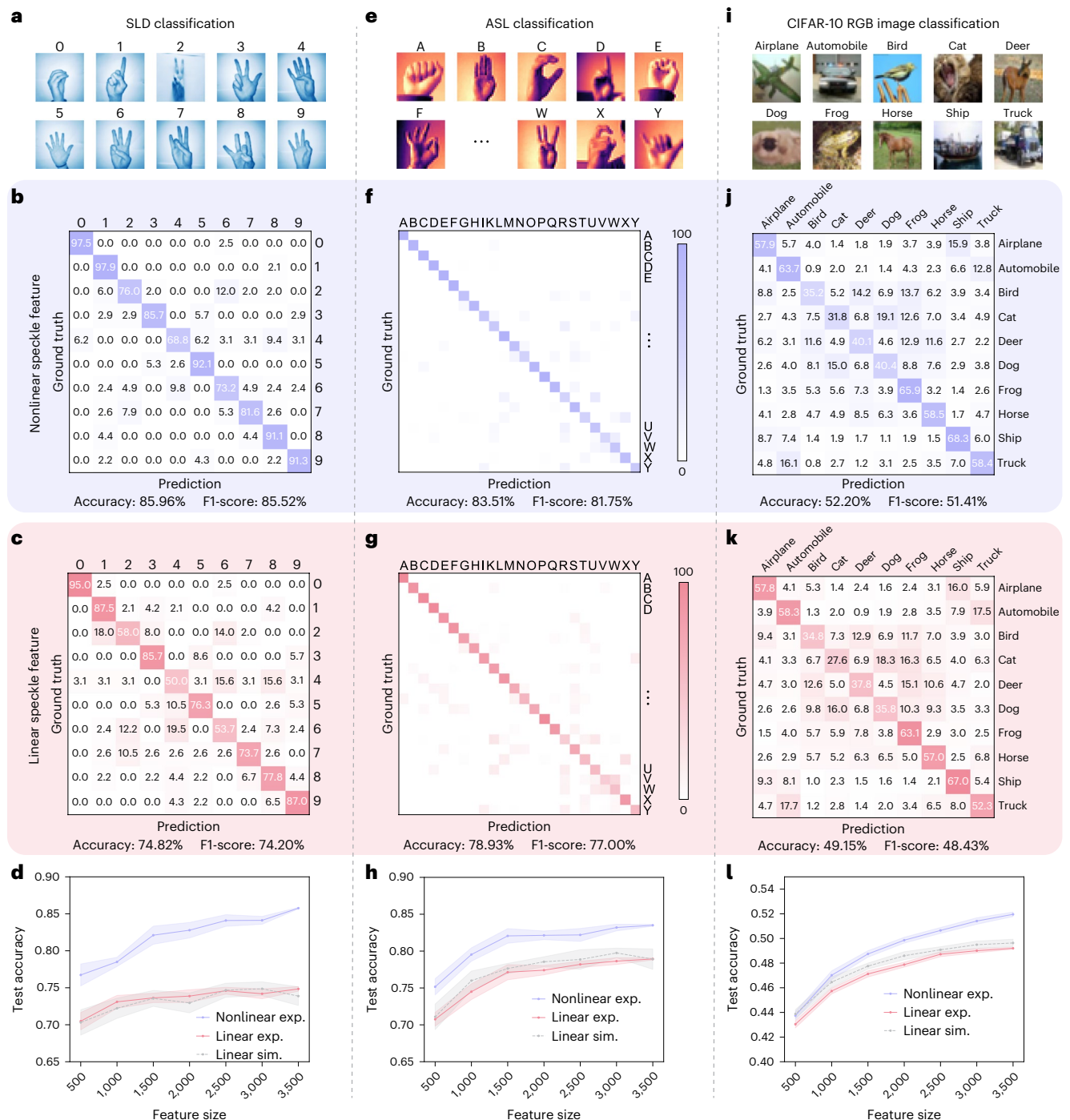


Fig. 2 | Speckle features for image classification. **a**, 10-class SLD images. **b, c**, The confusion matrices on the test set obtained by nonlinear (**b**) and linear (**c**) speckle features at a feature size of 3,500. Nonlinear and linear speckle features are used to classify three image datasets. **d**, Comparison of classification accuracies based on the experimental nonlinear feature (violet), experimental linear feature (red) and simulated linear feature (gray) at varying feature sizes. The nonlinear speckle feature outperforms its linear counterpart with an absolute accuracy increase ranging from 5.4% to 10.9%. **e**, 24-category ASL

alphabet images. **f, g**, Same as **b** and **c** for ASL alphabet dataset, respectively. **h**, Same as **d**, with test accuracy improvement from 4.0% to 5.0% using the nonlinear feature. **i**, CIFAR-10 RGB images. **j, k**, Same as **b** and **c** for the CIFAR-10 dataset, respectively. **l**, Same as **d**, with test accuracy improvement from 0.7% to 2.8% using the nonlinear feature. In **d**, **h** and **l**, the shaded area represents one s.d. from the mean value calculated from ten repeated tests. SLD is represented for **a–d**, ASL for **e–h** and CIFAR-10 for **i–l**.

database⁵³. Here we attain an accuracy improvement of approximately 3.1% (52.20% versus 49.15%) at a feature size of 3,500 (Fig. 2j, k). In addition, we also tackle another challenging image recognition task, the STL-10 database⁵⁴ that features a large input dimension of 27,648. The experimental results for classifying STL-10 (Supplementary Fig. 3) and CIFAR-10 (Fig. 2l) at various feature sizes show similar

behaviors and indicate the importance of incorporating nonlinearity for improved performance.

Univariate and multivariate regression

Our photonic neural system can also be adapted for regression tasks, as shown in Fig. 3. Different from the image classifications where the

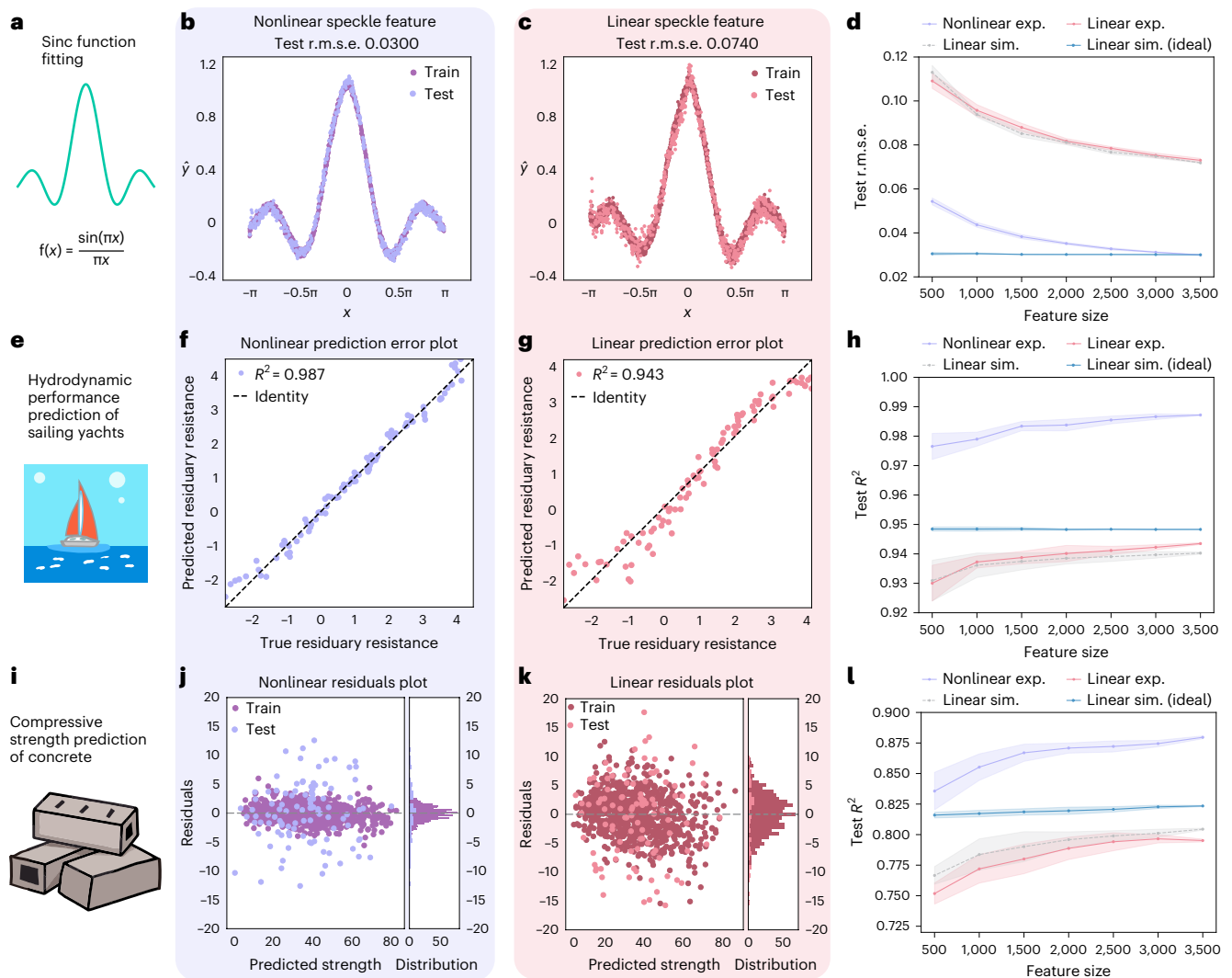


Fig. 3 | Speckle features for regression. **a**, Interpolation of a sinc function $f(x) = \sin(\pi x)/(\pi x)$ for $x \in [-\pi, \pi]$. **b, c**, The train and test sinc interpolation obtained by nonlinear (**b**) (test r.m.s.e. of 0.0300) and linear (**c**) (test r.m.s.e. of 0.0740) speckle features at a feature size of 3,500. **d**, Comparison of the test set regression errors based on the experimental nonlinear feature (nonlinear exp., violet), experimental linear feature (linear exp. red) and simulated linear features with (linear sim.) and without noise (linear sim. (ideal); dotted gray and solid blue, respectively) at varying feature sizes. **e**, Prediction of the hydrodynamic performance of sailing yachts based on their dimensions and velocity (six input parameters). **f, g**, Plots of prediction versus ground truth obtained by nonlinear (**f**) (test R^2 of 0.987) and linear (**g**) (test R^2 of 0.943) speckle features at a feature size of 3,500. **h**, Same as **d**, but the metric used is the R^2 . **i**, Prediction of the

concrete compressive strength based on its composition and age (eight input parameters). **j, k**, The train and test residual error distributions obtained by nonlinear (**j**) and linear (**k**) speckle features at a feature size of 3,500. **l**, Same as **h**. In **d, h** and **l**, the experimental nonlinear feature (violet) demonstrates improved performance compared with its linear counterpart (red) across all three regression tasks, surpassing even the performance achieved with ideal linear feature (solid blue) in multivariate regressions. The shaded regions show one s.d. above and below the mean of ten repeated experiments. The nonlinear and linear speckle features are used for three regression tasks: sinc interpolation for **a–d**, hydrodynamic performance prediction of yachts for **e–h** and compressive strength prediction of concrete for **i–l**.

system makes a decision based on the maximum likelihood, regression necessitates directly aligning with precise values. Therefore, regression tasks are more vulnerable to the influence of quantization and noise. In our system, both the SLM encoding and camera readout have 8 bit quantization, and the system's signal-to-noise ratio (SNR) is measured to be around 14 dB, with the noise dominated by the flickering of the SLM, laser fluctuation, temperature fluctuation and mechanical instability of the nonlinear scattering sample and optical elements⁵⁵ (Supplementary Note 2). A canonical regression example is to fit a nonlinear sinc function $y = \sin(\pi x)/(\pi x)$, where $x \in [-\pi, \pi]$ (Fig. 3a–d). We encode each input variable x to the SLM and record generated speckle features (see Methods for data processing). To fit the corresponding y value, these features must exhibit high nonlinearity regarding x . For such regression, we note that the phase encoding

nonlinearity plays an important role, as a larger encoding range provides higher-order nonlinear terms (Supplementary Note 3). An encoding range of $[0, 2\pi]$ is used in Fig. 3a–d. Figure 3b,c shows the experimental results of sinc interpolation based on 3,500 nonlinear and linear speckle features, respectively. Figure 3d illustrates their test root mean square errors (r.m.s.e.) at different feature sizes, together with the linear speckle simulation with and without considering the experimental SNR (denoted as 'Linear sim.' and 'Linear sim. (ideal)', respectively). It is evident that nonlinear speckle features achieves lower r.m.s.e., owing to the higher-order nonlinear components embedded within them. We also repeat the experiment above with different phase encoding ranges and find out that the nonlinear speckle gives lowest r.m.s.e. of 0.0069 under a range of $[0, 4\pi]$ (Supplementary Note 3).

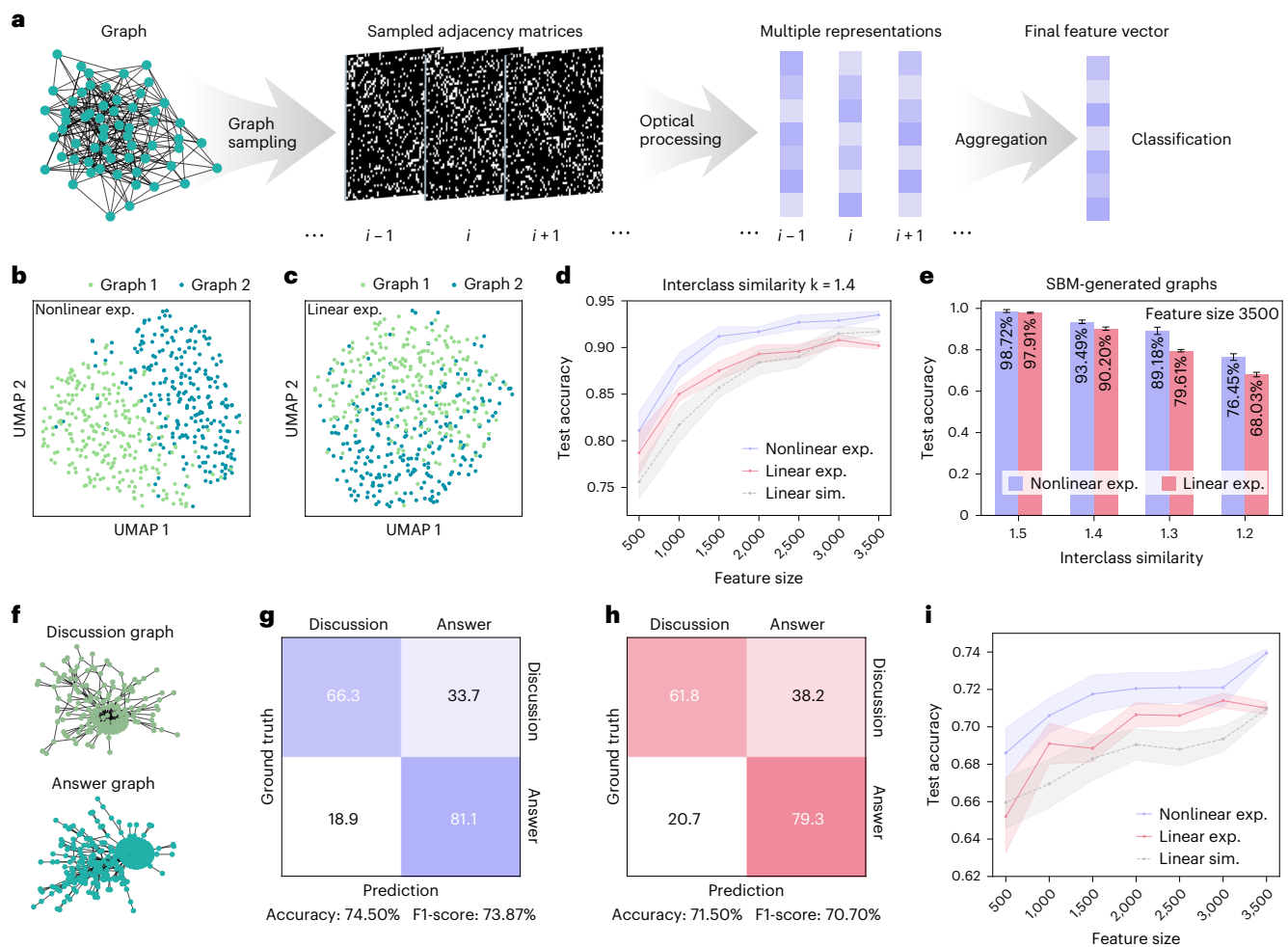


Fig. 4 | Speckle features for graph classification. **a**, General graph processing workflow. Multiple adjacency matrices are sampled from a graph, and their corresponding speckle features are optically extracted in our photonic neural system. The average of these speckle representations is taken as the final feature vector for classification. **b**, **c**, Visualizations of SBM-generated graph data (interclass similarity $k = 1.4$) with a UMAP obtained by nonlinear experimental (**b**) and linear experimental (**c**) speckle features at a feature size of 3,500. **d**, Comparison of classification accuracies based on the experimental nonlinear feature (nonlinear exp., violet), experimental linear feature (linear exp., red) and simulated linear feature (linear sim., gray) at varying feature sizes. The accuracy

gain varies from 2.1% to 3.7% for $k = 1.4$. **e**, The bar chart of mean test accuracies using nonlinear and linear features at varying interclass similarities. **f**, Illustration of discussion-based and question/answer-based graph samples from the RB dataset. **g**, **h**, The confusion matrices on the test set obtained by nonlinear (**g**) and linear (**h**) features at a feature size of 3,500. **i**, Same as **d**, with test accuracy improvement from 0.7% to 3.4% using the nonlinear feature. The shaded regions in **d** and **i** and the error bars in **e** show one s.d. above and below the mean of ten repeated experiments. The nonlinear and linear speckle features are used to classify two graph datasets: the synthetic graphs generated by the SBM for **b–e** and RB graphs for **f–i**.

In addition to univariate regression, we also explore two real-world multivariate applications. In Fig. 3e–h, we predict the hydrodynamic performance of sailing yachts based on their basic hull dimensions and the boat velocity⁵⁶. Figure 3f,g shows the regression results based on nonlinear and linear speckles at a feature size of 3,500. We employ the R^2 metric to quantify the regression performance (see Methods for metrics), and observe an improved R^2 using nonlinear speckle features (0.987 versus 0.943). Similarly, in another regression task that predicts the compressive strength of concrete from its composition and age⁵⁷ (Fig. 3i–l), the nonlinear speckle gives a residual distribution that is closer to zero, again indicating better performance. Furthermore, for these two tasks, the experimental nonlinear features even surpass the ideal linear random features simulated without noise (Fig. 3h,l).

Graph classification

Unlike Euclidean data such as images, many real-world objects are often described by their connections to other things and can be represented as a graph with nodes and edges, such as social networks. Numerous

frameworks have been developed to learn representations from non-Euclidean graph data, notably including frequent subgraph-based algorithms⁵⁸, graph convolutional networks⁵⁹ and graph kernels-based algorithms⁶⁰. Among them, graphlet kernel-based graph classification involves leveraging the frequencies of small, connected subgraphs (graphlets) sampled from the original graph to quantify structural similarity and enable classification tasks⁶¹. Interestingly, graph processing has recently been realized in optical linear diffractive units⁶² and linear light scattering⁶³. Inspired by the graphlet kernel methods with random features⁶³, we propose to extract graph information based on the nonlinear scattering medium. In our system, for moderate-sized graphs with similar number of nodes across graphs, we directly encode the adjacency matrix of each graph to the SLM and optically extract representations. As shown in Fig. 4b–e, we test our method on a synthetic dataset composed of two-category graphs generated by the stochastic block model (SBM)⁶⁴. We fix the node number of all graphs constant (60 in this study) and control the difficulty of graph classification via the interclass similarity k (closer to 1 indicates a higher similarity between

the two classes). To better visualize the superiority of nonlinear speckle features, we exploit the uniform manifold approximation and projection (UMAP) technique to reduce feature dimensions from 3,500 to 2 (Fig. 4b,c). Indeed, at a setting of $k = 1.4$, the nonlinear embeddings render better clustering of the two classes, leading to a higher test accuracy (Fig. 4d). Similar accuracy trends are observed with varying interclass similarities, as summarized in Fig. 4e. Since the rearrangement of nodes can alter a graph's adjacency matrix, the method used for SBM-generated graphs may not preserve the permutation invariance as graph convolutional networks⁶³. Therefore, we posit that the successful classification in this case is achieved by extracting the density of edges in graphs.

Moreover, we also address a real-world Reddit-binary (RB) graph database⁶⁵ (Fig. 4f–i). These graphs vary in size, from as small as six nodes and four edges to as large as 3,782 nodes and 4,071 edges, thus posing challenges for graph data encoding. To this end, we develop a generalized graph processing workflow, as shown in Fig. 4a. Given a large graph, we first sample multiple subgraphs with a uniform number of nodes and extract the adjacency matrix of each graph⁶³. The corresponding adjacency matrices of these graphlets are then divided into groups, and each group is subsequently encoded to the SLM altogether to accelerate the feature extraction process. Consequently, multiple subgraph representations are obtained and aggregated as a final feature vector of the graph via an average operation, before the digital classifier (see Methods for data processing). Our goal here is to discern between 'discussion-based' and 'question/answer-based' graphs (Fig. 4f). Again, the nonlinear representations show better classification results and reaffirm the added value of nonlinearity as shown in Fig. 4g–i (see Supplementary Table 3 for more digital algorithm baselines).

Computation analysis

To unveil the origin of optical nonlinearity superiority summarized in Supplementary Table 1, we first analyze the nonlinear optical computation based on the mathematical model presented in Fig. 1c. The SLD dataset with the image size of 64^2 is considered as an example. Different from linear speckle simulation (see Methods for data processing), simulating the nonlinear optical mapping would require a massive matrix W_n with $3,500 \times (64^2)^2 = 58,720,256,000$ entries, equivalent to 437.5 GB of memory usage in a digital hardware for an output feature size of 3,500 (Supplementary Note 4). Matrix–vector multiplication at such an extreme scale becomes computationally intense and memory hungry. In our analog system, the computations are optically performed without measuring and storing the matrix, estimated as 18.8 T floating point operations per second, albeit the matrix is fixed (Supplementary Note 4). To examine the role of nonlinearity in a simulation-friendly manner, we consider a much simplified nonlinear optical model if we assume only scalar fields instead of vector fields (Supplementary Note 1 and Supplementary Fig. 4). The simplified nonlinear simulation results also outperform the experimental linear features, yet fall behind the experimental nonlinear results in Fig. 1d, indicating richer computations beyond a two-layer NN are achieved in the experiments. Indeed, the simplified simulation does not account for the full second-order nonlinear tensor and randomized orientations and sizes of LN nanocrystals⁶⁶ (Supplementary Note 1). To illustrate the computing capability of the photonic neural hardware, we also benchmark our system with extensive simulations of several well-known digital baselines (Supplementary Note 5 and Supplementary Tables 2–5).

Discussion

Despite the improved performance of our photonic neural system, using a table-top femtosecond laser in this study compromises the energy efficiency. Noticeably, sum-frequency speckles have been generated in random assembly of the same LN nanocrystals with CW illumination⁶⁷. Obtaining CW-pumped SH speckles from our platform

is an exciting perspective, as it would dramatically reduce the needed computation power, and the total power consumption in optics can be within tens of watt considering the laser, SLM and camera used for experiments (Supplementary Note 4). While our system is not best suited to implement isomorphisms of digital NNs (for example, optical transformers¹⁴), the efficient physical transformation performed by the LN slab can also be employed as a building block within a deep NN². By incorporating trainable parameters, either from the analog or digital domains, more sophisticated ML tasks can be investigated using backpropagation-free training techniques⁶⁸. Although using a multilayer digital backend on speckle features can improve the performance (Supplementary Table 7), it compromises the training simplicity and we use one layer digital readout throughout the study. While this work randomly draws speckle features from measured camera images, searching 'superior' features via feature selection methods could reduce digital computational overhead. Moreover, the use of $\chi^{(2)}$ nonlinear media for computation also opens up possibilities for quantum or quantum-enhanced computing with entangled photons⁶⁹. Given the extensive dimensionality and versatility of our system, we foresee it as a powerful analog computing platform potentially serving as a building block of NNs for a wide range of applications.

Methods

Sample information

We assemble the slabs starting from crystalline LN nanoparticles that are produced via solvothermal synthesis. The precursors oxides Nb_2O_5 (HC Starck, 99.92%) and LiOH (Aldrich, 98%) are dispersed in a mixture of ethylene glycol and distilled water. After ultrasonication, the suspension is poured into a polytetrafluoroethylene-coated stainless steel acid digestion bomb (model PA4748, volume 120 ml, Parr Instrument Company) and hydrothermally treated at 250 °C for 70 h. Afterwards, the reaction product is washed with water by centrifugation. This chemical synthesis allows to precisely control the size distribution of the produced nanoparticles. The nanoparticles have sizes from 100 to 400 nm, a linear refractive index of $n \approx 2.3$ and negligible absorption at visible wavelengths⁷⁰. They have a non-centrosymmetric hexagonal R3c crystal structure that enables SH generation. We assemble the nanoparticles into slabs by drop deposition and solvent evaporation. An aqueous suspension (1 wt %) of the LN nanoparticles is mixed with a ratio of 60:1 polyvinyl alcohol and deposited over a glass substrate framed with hydrophilic tape. The sample is placed onto a horizontal substrate holder and kept at 0 °C for 24 h. At the end of the process, the nanoparticles are deposited and the water evaporated. The slabs have a subwavelength transport mean free path, which results in multiple light scattering within the thickness of 5 μm (measured by profilometry). A thorough explanation of the fabrication process and further characterization of the linear and nonlinear optical properties of the slabs is reported elsewhere⁴¹.

Experimental setup

The detailed experimental setup is shown in Supplementary Fig. 5. A mode-locked Ti:sapphire laser (MaiTai HP, Spectra-Physics) is used as the pump, producing an 80 MHz repetition-rate, 100 femtosecond laser pulses at a central wavelength of 800 nm. The laser beam goes through a half-wave plate and a polarizing beam splitter to align the beam polarization to the working axis of the SLM as well as to adjust the light intensity. A spectrometer (Ocean optics, HR4000) collects a small portion of light to monitor the emission spectrum of the laser. The laser beam is expanded by a 4– f system via two lenses ($L_1, f = 12$ mm; $L_2, f = 100$ mm) and then illuminates a reflective phase-only SLM (HSP512L-1064, Meadowlarks). The datasets are encoded to the spatial wavefront of the light via the SLM, which is relayed by another 4– f system ($L_3, f = 150$ mm and $L_4, f = 150$ mm) to the back focal plane of an objective lens (obj. 1, Plan N 20 $\times$ 0.4, Olympus). As such, the pump is delivered to the scattering LN slab via obj. 1, and the scattered pump

and the generated SH light are collected by an identical objective lens obj. 2. A dichroic mirror (DMSP650r, Thorlabs) is used to separate the FH and SH components. Both FH and SH speckles are captured by their respective cameras (Manta, G-046, Allied Vision) via tube lenses (L5, $f = 100$ mm and L6, $f = 150$ mm). In the FH path, a linear polarizer (LP1, LPVIS100-MP, Thorlabs) and neutral density filters are used to select the beam polarization vertical to the incident light and attenuate the FH intensity, respectively. While in the SH path, a linear polarizer (LP2, LPUV100-MP2, Thorlabs) is utilized to select linear polarized SH, and a bandpass filter (FBH400-40, Thorlabs) is also employed to completely suppress the residual pump in the SH path. In the experiment, the FH and SH speckles are measured separately due to the dispersion of the collecting objective lens. The overall experimental system stability is discussed in Supplementary Note 2.

Data processing in experiments and simulations

We employ a phase-only SLM to encode input data into the optical fields, with a phase encoding range of $[0, \pi]$ unless otherwise specified. Before data encoding, we use minimum–maximum normalization to linearly scale the original data to be within the range of $[0, 1]$. We use the central region of the SLM (around 400×400 pixels) for data encoding, while the periphery pixels are blocked by the aperture of the objective lens. Different macropixel sizes are used depending on different data dimensions (Supplementary Table 1). In cases the central region is not fully used, the unmodulated pixels act as a static bias. For grayscale image datasets, such as SLD, ASL, MNIST and FashionMNIST, we directly display each input image on the SLM. While for color image datasets, such as CIFAR-10 and STL-10, we concatenate three channels of each image (red, green and blue) together with its grayscale version as four blocks of a single composite image. As for univariate regression, the input variable cannot be directly encoded on the SLM, because a uniform phase change will not alter the generated speckle features. Instead, we employ a fixed, small-scale linear random matrix (30×30) and encode the product of the input variable and the matrix to the SLM, as has been implemented previously⁴⁰. Different phase encoding ranges are investigated (Supplementary Note 3), and this is the only case we explore encoding range other than $[0, \pi]$. In contrast to univariate regression, the input variables are directly encoded on the SLM in multivariate regression with proper macropixel sizes. For SBM-generated and Reddit-binary graph datasets, we encode the adjacency matrix of the graph to the SLM. In this study, we focus solely on undirected graphs and binary adjacency matrices with only connection information, though the proposed system is not limited to such cases. All SBM graphs consist of 60 nodes, therefore, the dimensions of their corresponding adjacency matrices are 60×60 . Those adjacency matrices are directly displayed on the SLM. For the RB database, we first use the random walk method to sample each original graph with a sample size of six vertices for 2,000 times. Subsequently, we group the 2,000 sampled adjacency matrices into 20 distinctive phase masks, each comprising 100 matrices. These phase masks are loaded sequentially to the SLM to derive 20 speckle representations, which are then averaged to form as the final graph representation used for following digital readout.

To extract optical features, we downsample the measured speckle images with a stride size matching the speckle grain size. The latter is estimated by speckle autocorrelation. Therefore, distinctive and roughly independent optical features are obtained and are further flattened as a feature vector (feature size ~3,500 for both FH and SH speckles). Feature vectors with smaller feature sizes are randomly drawn from the complete feature vectors.

The optically generated feature vectors are then fed into a digital readout layer for ML tasks. It performs a simple linear regression trained by the Tikhonov regularization method. To do so, we organize the N -dimensional feature vectors of K samples into a matrix X . The objective is to find a linear combination of the feature vector elements that can best predict each label (grouped in a vector y). In other words,

we aim to adjust the weight matrix W_{out} to minimize the ℓ_2 norm of the residual, that is, $\|XW_{\text{out}} - y\|_2^2$. To penalize large weight values, we add an additional regularization term $\beta \|W_{\text{out}}\|_2^2$ to the objective function. This gives the analytical solution of the readout matrix as $W_{\text{out}} = (X^T X + \beta I)^{-1} X^T y$. An optimal regularization parameter β can enhance the model's generalization ability or reduce the risk of overfitting, particularly when $K < N$. As such, we search the best β in all cases for fair comparisons. Note that when we explore the nonlinear advantages, for two optical linear and nonlinear speckle feature vectors, we keep their dimensions the same and train their respective digital back-end networks given a task. The overall computing processes of the optical linear and nonlinear paths are also summarized in Supplementary Fig. 1 in a step-by-step manner with examples.

To simulate the computing performance of speckle features, we initialize random Gaussian independent and identically distributed complex matrices, whose dimensions are determined by the input data size and output feature size. For linear speckle simulation, we employ a single random matrix in the digital model, rather than multiple of them corresponding to a multispectral transmission matrix⁴⁵. This is valid, as we observe the computing performance in these two cases are very similar, while a single random matrix provides a straightforward baseline same as linear RP using CW light. Compared with image and graph classification tasks, we notice that the regression performance is more sensitive to quantization and noise. Therefore, as shown in Fig. 3 of the main text, we incorporate the quantization and SNR level (according to the experimental condition) when simulating the regression tasks. The effect of SNR is emulated by adding white Gaussian noise to the speckle intensity. Throughout all experimental data analysis and simulations, we use both MATLAB software (MathWorks) and PyCharm software (JetBrains) on two computers: one equipped with an Intel(R) Core(TM) i7-6700 central processing unit (CPU) and 32 GB random-access memory (RAM) and another with an AMD EPYC 7351P CPU, two NVIDIA GeForce RTX 2080 Ti graphics processing units (GPUs) and a 64 GB RAM.

Performance evaluation metrics and visualization techniques

Here we describe metrics and visualization methods used in this work for ML data analysis and evaluation. In classification tasks, 'confusion matrix' provides a comprehensive summary of classification model performance, showing the number of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions. In this work, each element of the confusion matrix indicates the percentage of predictions relative to the total number of instances in that category. 'Accuracy' quantifies the overall correctness of a classification model, expressing the proportion of correct predictions over the total number of predictions, that is, $\text{accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$. The 'F1-score' combines

precision = $\frac{TP}{TP+FP}$ and recall = $\frac{TP}{TP+FN}$ and is defined as the harmonic mean of precision and recall, that is, $\text{F1-score} = \frac{2 \cdot \frac{TP}{TP+FP} \cdot \frac{TP}{TP+FN}}{\frac{TP}{TP+FP} + \frac{TP}{TP+FN}} = \frac{2TP}{2TP+FP+FN}$.

It offers a balanced measure of model performance, especially when dealing with imbalanced datasets.

For regression tasks, r.m.s.e. = $\sqrt{\frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{N}}$ is a widely used metric to quantify the average difference between prediction $\hat{y}$ and ground truth y . Similarly, $R^2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$ measures how well the model's

prediction fits the actual data, with a higher value indicating a better fit. The error plots visually display the relationship between model predictions and actual targets. By comparing this plot against the 45-degree line ($\hat{y} = y$, where the prediction perfectly matches the ground truth), it facilitates a clear visualization of a model's variance in a given task. Residual plots graphically represent the differences between ground truths and predicted values, giving insights into model

accuracy and potential patterns in the errors. The residual plot from a well-fitted regression exhibits a dense cluster of points near the origin, accompanied by a sparse and unbiased distribution away from it.

UMAP is a dimension reduction technique widely employed for visualizing high-dimensional data in a lower-dimensional space, while preserving important relationships and structures of data. In brevity, UMAP initially constructs a weighted, high-dimensional graph, wherein each vertex corresponds to a feature vector and each edge weight denotes the likelihood that two vertices are connected. Then UMAP refines a low-dimensional layout to make it as similar to the initial graph as possible.

Data availability

The raw datasets before optical processing are all publicly available. Specifically, the raw data for SLDs dataset is downloaded from ref. 47, ASL alphabet dataset is downloaded at <https://www.kaggle.com/datasets/datamunge/sign-language-mnist>, CIFAR-10 dataset is downloaded from ref. 53, yacht hydrodynamics dataset is downloaded from ref. 56, concrete compressive strength dataset is downloaded from ref. 57, graph dataset generated by the SBM is proposed by ref. 64, Reddit-binary graph dataset is downloaded from ref. 65, MNIST dataset is downloaded from ref. 49, FashionMNIST dataset is downloaded from ref. 50 and STL-10 dataset is downloaded from ref. 54. The recorded experimental speckle feature datasets are available at <https://doi.org/10.5281/zenodo.10799862> and <https://doi.org/10.5281/zenodo.8392103> (refs. 71,72). Source data for Figs. 2, 3 and 4 are provided with this paper.

Code availability

The code used to produce the results within this work is openly available at <https://doi.org/10.5281/zenodo.10799862> (ref. 71).

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Author contributions

J.H. and S.G. conceived the project. H.W. and J.H. developed the experimental setup with the assistance of F.X. H.W. and J.H. performed the experiments and simulations and processed and analyzed the results. A.M., A.N., X.L., R.S. and R.G. fabricated and characterized the LN samples and provided insights of the physical model. H.W. and J.H. wrote the paper with input from all authors. S.G., R.G., Q.L. and J.H. supervised the project.

Competing interests

The authors declare no competing interests.

Additional information

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